DIPARTIMENTO DI ELETTRONICA, INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)
September 26, 2016

Name:
Surname:
Student ID:
Row: Column:

Time: 2 hour 30 minutes Prof. Marcello Restelli Maximum Mark: 31

- The following exam is composed of **10 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided outside these spaces will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices** like laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 31

Exercise 1 (5 points)

Describe the supervised learning technique denominated Support Vector Machines (SVMs) for classification problems.

Exercise 2 (5 points)

Describe the UCB1 algorithm for multi-armed bandit problems.

Exercise 3 (5 points)

Describe the TD(λ) algorithm for learning the value function of a given policy.

Exercise 4 (2 point)

Consider separately the following characteristics for a Machine Learning problem:

1. Reduced computational capabilities;
2. Having *a priori* information about the problem;
3. Availability of huge amount of data;
4. Learning in a real–time scenario.

Provide motivations for the use of either a **parametric** or **non–parametric** method.

1. PARAMETRIC/NON–PARAMETRIC: if we are able to perform the training phase on a different device, a parametric model would be a good choice in the case we have reduced computational capabilities. If we do not have this possibility a non–parametric method would be a good choice since it does not require any training;
2. PARAMETRIC: *A priori* information can be easily included in the model you are using to solver the specific problem, for instance by considering a Bayesian method and considering a prior distribution over the model parameters;
3. PARAMETRIC: if we resort to a non–parametric method we will have that the evaluation phase for a newly coming sample will have a computational complexity proportional to the number of data, while parametric models would require only an amount of time proportional to the number of parameters;
4. PARAMETRIC/NON–PARAMETRIC: since usually the training phase is costly and takes a long time, it would not be a good idea to use parametric methods to provide a prediction in a limited amount of time. If we are able to provide an on–line training method, which is able to incrementally acquire information on newly seen data, we could also use a parametric method.

Exercise 5 (2 point)

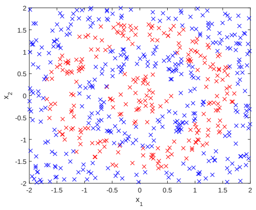
Which of the following statements about the Lasso regularization are true or false. Provide motivations for your answers.

1. The Lasso regularization selects automatically a set of variables that are meaningful in the considered problem;
2. The Lasso penalty term helps to decrease the bias of the considered model;
3. The minimum of the loss function of a linear regression problem with Lasso penalty has closed form solution;
4. The use of a Lasso regularization term increases the sparsity of the model.

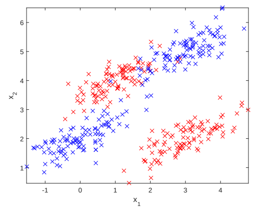
1. TRUE: since the effect of lasso regularization is to put some of the coefficient to zero, it automatically selects the set of variables which are more meaningful for the problem;
2. FALSE: the effect of a regularization term is to simplify the model (reduce the variance), thus avoiding overfitting effects, which clearly might increase the model bias;
3. FALSE: while the classical linear regression has a closed form solution (the LS one), in the case of lasso we have a problem which is not differentiable when w.r.t. the parameters (the module is not differentiable in zero);
4. TRUE: since the effect is to put some parameters to zero we will get a sparse model in the end.

Exercise 6 (2 point)

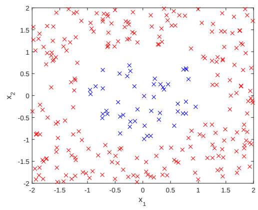
For which one of the following datasets would you use the kernel trick to represent your data? Would you use some other methodology? Provide motivations for your choices.



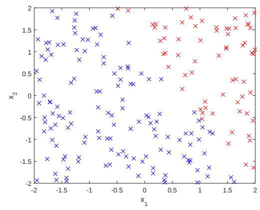
(a)



(b)



(c)



(d)

a) KERNEL: even the transformation of the problem in polar coordinates does not allow us to find a linear separating hyperplane, thus the use of a kernel is suggested;

b) KERNEL: the structure of the problem and the non linear separability suggests the use of Gaussian kernel;

c) NO KERNEL: the transformation of the problem in polar coordinates allow us to separate linearly the two classes, thus the use of a kernel would be unnecessary;

d) NO KERNEL: the classes are linearly separable.

Exercise 7 (2 point)

Categorize the following Machine Learning problems:

1. Online advertisement display selection;
2. Predict the danger level in a specific area;
3. News aggregation;
4. Decide a strategy for stock trading.

For each one of them suggest a set of features that might be useful to solve the problem and a method to solve it (e.g., Iris flower identification problem is a classification problem since there is no topology on the output space, logistic regression could be used to solve it by considering as features petal length and sepal width).

1. MAB, UCB1, number of click as reward and ads as actions;
2. Regression (ordered outcome), Linear regression, number of chemical industries in the area, earthquake chance;
3. Clustering, K–means, word counting statistics;
4. RL, Q–learning, increase and decrease of the stock value as reward and buy or sell as actions.

Exercise 8 (2 point)

After performing Ridge regression on a dataset with $\lambda = 10^{-4}$ we get one of the following sets of eigenvalues Λ for the matrix $(\Phi^T\Phi + \lambda I)$:

1. $\Lambda = \{-0.032,\ 0.000245,\ 5\}$;
2. $\Lambda = \{0.00000245,\ 0.014,\ 1\}$;
3. $\Lambda = \{0.000245,\ 0.000245,\ 5\}$;
4. $\Lambda = \{0.000245,\ 0.022,\ 5\}$;

Explain whether these sets are plausible solutions or not.

The eigenvalues of a regularized matrix with Ridge term should be all greater or equal than λ, thus only the options 3 and 4 are plausible.

Exercise 9 (3 points)

Consider the MDP in Figure 1 with transition probabilities $\alpha = 0$ and $\beta = 0.5$, discount factor $\gamma = 1$, instantaneous rewards $r_{search} = 4$ and $r_{wait} = 1$ and the following policy:

$$\pi(high, search) = 0.5$$
$$\pi(high, wait) = 0.5$$
$$\pi(low, search) = 1$$

1. Compute the Value function in the case the MDP stops after two steps.
2. Compute the action–value function for each action value pair in the case the MDP stops after a single step.

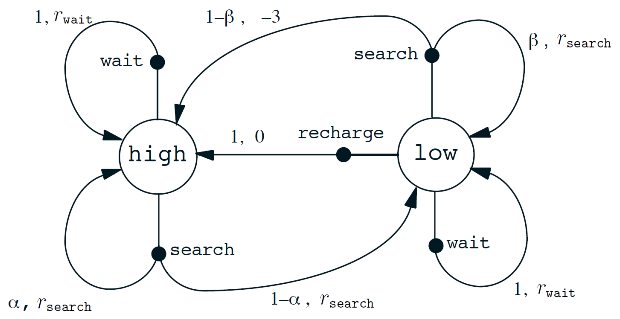


Figure 1: The MDP for the cleaning robot.

1.

$$V^{\pi}(s) = \sum_a \pi(s|a)\left[R(s,a) + \gamma \sum_{s'} P(s'|s,a)V^{\pi}(s')\right]$$
$$= \sum_a \pi(s|a)\left[R(s,a) + \sum_{s'} P(s'|s,a)\sum_{a'} R(s',a')\right]$$

$$V(high) = 0.5\left[4 + 1\left(0\sum_{a'} R(high,a') + 1\sum_{a'} R(low,a')\right)\right] + 0.5\left(1 + \sum_{a'} R(high,a')\right)$$
$$= 0.5\,(4 + 0.5\cdot(-3) + 0.5\cdot 4) + 0.5(1 + 0.5\cdot 1 + 0.5\cdot 4) = 0.5\cdot 4.5 + 0.5\cdot 3.5 = 4$$

$$V(low) = 1\left[0.5 + 1\left(0.5\sum_{a'} R(high,a') + 0.5\sum_{a'} R(low,a')\right)\right]$$
$$= 0.5 + 0.5\cdot 2.5 + 0.5\cdot 0.5 = 2$$

2.

$$Q(high, search) = 0.5\cdot 4 + 0.5\cdot 4 = 4$$
$$Q(high, wait) = 1$$
$$Q(low, search) = 0.5\cdot(-3) + 0.5\cdot 4 = 0.5$$
$$Q(low, wait) = 1$$
$$Q(low, recharge) = 0$$

Exercise 10 (3 points)

Consider a clustering algorithm $\mathcal{C}$ that, given K, the number of clusters, and N, the number of observations, try all possible assignments of the N observations into K clusters. Then, select one of the assignments that minimizes:

$$J = \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} ||x_i - \mu_k||_2^2$$

where x_i is the i–th sample, r_{ik} is a binary term which specifies if the sample x_i is in the cluster with index k and μ_k is the centroid of the k–th cluster.

1. In terms of N and K, how many potential solutions will your algorithm try?;
2. Provide the complexity of the K–means algorithm in terms of N and K .
3. Discuss the advantages and drawbacks of the $\mathcal{C}$ algorithm and the K–means.

Provide motivations for your answers.

1. $O(K^N)$ since we have K independent choice for each one of the N points;
2. $O(KN)$ since at each iteration we have to compute K distances from the cluster centers for each one of N points;
3. $\mathcal{C}$ has the advantage to find the configuration minimizing J, while K–means might get stuck in a local optimum. K–means is less computationally demanding than $\mathcal{C}$.